ON SELF-ORGANIZED DYNAMICS WITH FINITE SPEED OF INFORMATION PROPAGATION AND PROCESSING DELAY

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Abstract. We introduce and study a modified Cucker-Smale model where information between agents propagates with significantly large speed compared with the agents' speeds. This leads to a system of ordinary differential equations with delay. We prove an existence and uniqueness result for the associated initial-value problem and illustrate the behavior of the solutions numerically.

1. Introduction

Self-organized dynamics refers to the emergence of coherent macroscopic behavior from relatively simple interactions among many individual agents, without centralized coordination. Examples range from flocking birds and schooling fish to pedestrian motion, opinion formation, distributed robotic systems, and autonomous vehicle coordination. One of the most influential mathematical frameworks for studying this phenomenon is the Cucker–Smale model [2], which is an important tool for explaining self-organized behaviors in complex systems. These emerging behaviors are typically described by the so-called flocking solutions. The central question in [2] is deceptively simple: under what conditions can local pairwise interactions drive a population of independently moving agents toward global consensus? The importance of the Cucker–Smale classical model and subsequent related contributions by many authors lies less in their literal description of biological flocks than in the mathematical paradigm it introduced. These models connect ordinary differential equations, dynamical systems, graph theory, kinetic theory, nonlinear PDEs, probability, optimal control, and numerical analysis. Recent research increasingly concentrates on connecting the microscopic particle equations to kinetic and macroscopic models, extending flocking to heterogeneous or restricted communication networks, and understanding stochasticity, singular interactions, control, delays, and several forms of clustering (see [1, 3, 5, 6, 8] and references therein for recent contributions and informative surveys).

A modified Cucker-Smale model was introduced by Liu and Wu in [7], where the authors incorporate processing delay in the influence function into the Cucker-Smale model. In what follows, $\mathbf{x}(t) = (\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)) \in \mathbb{R}^{Nd}$ and $\mathbf{v}(t) = (\mathbf{v}_1(t), \dots, \mathbf{v}_N(t)) \in \mathbb{R}^{Nd}$ denote the matrices whose columns consist of positions and velocities at any given time t, respectively, of N agents. More precisely, for a self-organized group of N agents, each agent i is characterized by its position $\mathbf{x}_i(t) \in \mathbb{R}^d$ and velocity $\mathbf{v}_i(t) \in \mathbb{R}^d$ at any given time t, where the space dimension d is a positive integer. Under these notations, the dynamic is governed by the

following system of ordinary differential equations

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i(t) \tag{1}$$

$$\frac{d\mathbf{v}_i}{dt} = \frac{\alpha}{N}\sum_{j\neq i}^{N} a_{ij}(t-\tau)(\mathbf{v}_j(t-\tau)-\mathbf{v}_i(t)), \tag{2}$$

where α measures the interaction strength and τ denotes the communication time between agents i and j, which includes the response time of agent i. Here, a_{ij} is either as in the Cucker-Smale formulation,

$$a_{ij}(t) = \frac{I(d_{ij}(t))}{N},$$

with $d_{ij}(t) := \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|_2$ the Euclidian distance between agents i and j at time t, or as in the more recent formulation by Motsch and Tadmor [9],

$$a_{ij}(t) = \frac{I(d_{ij}(t))}{\sum_{k=1}^{N} I(d_{ik}(t))},$$

where the so-called influence function I is a strictly positive monotonically decreasing continuous function, such as $I(r) = (1+r^2)^{-\beta}$, for $r \geq 0$ and β a positive constant. Notice that the coefficients a_{ij} depend on the components of $\mathbf{x}$.

It is shown in [7] that the processing delay leaves the qualitative flocking behaviors unchanged, but alters the flocking velocity in a complicated nonlinear manner, depending on both the initial velocity and the variation of the initial positions.

The paper is organized as follows: In Section 2 we introduce our modified Cucker-Smale model and provide the proof of local existence and uniqueness of solutions. In Section 3, we illustrate the behavior of solutions, with special emphasis on flocking behavior.

2. Finite information propagation speed and delayed influence

In this paper, we extend the model in [7] by assuming that the information propagates with a constant finite speed $\xi > 0$ such as the speed of sound or light. We also assume that the speeds of the agents (e.g., birds, bees, drones, etc) are much lower than the information speed ξ. It means that the distance between any two agents changes at very low rates compared to the constant information speed ξ. A sufficient condition is to assume that the speed of any agent is bounded from above by η, with η significantly smaller than ξ. This assumption allows us to consider that the agent i located at $\mathbf{x}_i = \mathbf{x}_i(t)$ at time $t > 0$ observes the agent j at time $t - \xi^{-1}d_{ij}(t)$. Recall that $d_{ij}(t)$ is the distance between the agents i and j at current time t.

With these assumptions, for $t \in [0, T]$, with $T > 0$, our model reads:

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i(t) \tag{3}$$

$$\frac{d\mathbf{v}_i}{dt} = \frac{\alpha}{N}\sum_{j\neq i}^{N} a_{ij}(t-\xi^{-1}d_{ij}(t))(\mathbf{v}_j(t-\xi^{-1}d_{ij}(t))-\mathbf{v}_i(t)), \tag{4}$$

while preserving the quantities defined in the model (1)-(2) introduced in [7] and overviwed above. Since (3)-(4) is a delayed model, corresponding initial conditions need to be specified

$$\mathbf{x}_i(t) = \mathbf{f}_i(t), \quad \mathbf{v}_i(t) = \mathbf{g}_i(t), \quad i \in [N] := \{1, 2, \ldots, N\}, \quad \text{for } t \leq 0, \tag{5}$$

where $\mathbf{f}_i$ and $\mathbf{g}_i$ are given continuous vector-valued functions, with $|\mathbf{g}_i(t)| \leq \eta$, $i \in [N]$. Henceforth, we also assume that the coefficients a_{ij} satisfy the following conditions on their domain of definition:

$$a_{ij}(t) \leq \frac{1}{N-1}, \quad \text{for all } t, \tag{6}$$

and are uniformly Lipschitz continuous, i.e., there exists a positive constant L such that

$$|a_{ij}(t) - a_{ij}(s)| \leq L|t - s|, \quad \text{for all } t \text{ and } s. \tag{7}$$

Note that we need to impose the obvious assumption that

$$\mathbf{f}_i(t) = \mathbf{f}_i(0) - \int_t^0 \mathbf{g}_i(s)ds, \quad i = 1, 2, \ldots, N, \quad \text{for } t \leq 0, \tag{8}$$

so that we only need to specify $\mathbf{g}_i(t)$ for $t \leq 0$ and $\mathbf{f}_i(0)$.

Naturally, in order to find the solution of (3)-(4), the values of the initial conditions (5) are only relevant on a bounded time interval $[-J, 0]$, with J to be specified in what follows.

Let $d_{\mathbf{x}}(t) := \max\limits_{1 \leq i,j \leq N} d_{ij}(t)$ and $d_{\mathbf{v}}(t) := \max\limits_{1 \leq i,j \leq N} \|\mathbf{v}_i(t) - \mathbf{v}_j(t)\|_2$. From our assumption regarding agents' speeds, notice that

$$d_{\mathbf{v}}(t) \leq 2\eta, \tag{9}$$

for all values of $t \geq 0$. Hence,

$$\begin{aligned}
d_{ij}(t) &= \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|_2 \\
&= \left\| \mathbf{x}_i(0) - \mathbf{x}_j(0) + \int_0^t (\mathbf{v}_i(s) - \mathbf{v}_j(s))ds \right\|_2 \\
&\leq \|\mathbf{x}_i(0) - \mathbf{x}_j(0)\|_2 + \left\| \int_0^t (\mathbf{v}_i(s) - \mathbf{v}_j(s))ds \right\|_2 \\
&\leq d_{ij}(0) + \int_0^t \|\mathbf{v}_j(s) - \mathbf{v}_i(s)\|_2 ds \\
&\leq d_{\mathbf{x}}(0) + \int_0^t d_{\mathbf{v}}(s)ds \\
&< d_{\mathbf{x}}(0) + 2\eta t \quad \text{(from (9))}
\end{aligned}$$

for all $t \geq 0$ and any choice of i and j. Thus,

$$d_{\mathbf{x}}(t) < d_{\mathbf{x}}(0) + 2\eta t$$

and

$$\begin{aligned}
t - \xi^{-1} d_{ij}(t) &\geq t - \xi^{-1} d_{\mathbf{x}}(t) \\
&\geq t - \xi^{-1}(d_{\mathbf{x}}(0) + 2\eta t),
\end{aligned}$$

for all $t \geq 0$. Therefore, in order to consider the solution of (1)-(2) on a bounded interval $[0, T]$, the values of the initial data (5) are necessary only on a bounded time interval $[-J, 0]$,

where $J = \xi^{-1}(d_{\mathbf{x}}(0)+2\eta T)$. Notice that J depends on T, ξ, η, and the agents' configuration at $t = 0$.

The first goal of this work is to prove (local) existence and uniqueness of solutions to the delay initial-value problem (3)-(5).

Theorem 1. *Under the above assumptions and notations, the delay initial-value problem (3)-(5) admits a unique solution on the interval $[-J, T]$, for $T > 0$ sufficiently small.*

Proof. For $\theta > \eta$ fixed for the moment, consider the following functional space

$$V := \{\mathbf{v} \in C([-J,T];\mathbb{R}^{Nd}) : \|\mathbf{v}_i(t)\|_2 \leq \theta,\ t \in [0,T],\ \text{and } \mathbf{v}_i(t) = \mathbf{g}_i(t) \text{ if } t \leq 0,\ \text{for all } i\},$$

where $\mathbf{v}(t) = (\mathbf{v}_1(t), \mathbf{v}_2(t), \ldots, \mathbf{v}_N(t)) \in \mathbb{R}^{Nd}$.

It is easy to observe that V is a Banach space with respect to the norm:

$$\|\mathbf{v}\| := \max_{t\in[-J,T]} \max_{i\in[N]} \|\mathbf{v}_i(t)\|_2.$$

Let $S : V \to V$ be the operator componentwisely defined by

$$S_i(\mathbf{w})(t) := \begin{cases} \mathbf{g}_i(t), & t \in [-J, 0], \\ \mathbf{g}_i(0) + \alpha \displaystyle\int_0^t \sum_{j\neq i}^{N} a_{ij}(s - \xi^{-1}w_{ij}(s))(\mathbf{w}_j(s - \xi^{-1}w_{ij}(s)) - \mathbf{w}_i(s))ds, & t \in [0, T], \end{cases}$$

where $w_{ij}(t) := \|\mathbf{w}_i(t) - \mathbf{w}_j(t)\|_2$ for $t \in [-J, T]$ and $i, j \in [N]$.

The goal is to prove that S maps the space V into itself and is a contraction with respect to $\|\cdot\|$.

For $\mathbf{w} \in V$, it is obvious that $S(\mathbf{w}) \in C([-J,T];\mathbb{R}^{Nd})$ and $S_i(\mathbf{w})(t) = \mathbf{g}_i(t)$, for all $t \leq 0$ and $i \in [N]$. Furthermore, we have

$$\|S_i(\mathbf{w})(t)\|_2 = \|\mathbf{g}_i(t)\|_2 \leq \eta < \theta, \quad \text{for } t \leq 0 \text{ and } i \in [N],$$

and

$$\begin{aligned} \|S_i(\mathbf{w})(t)\|_2 &\leq \|\mathbf{g}_i(0)\|_2 + \alpha \int_0^t \sum_{j\neq i}^{N} |a_{ij}(s - \xi^{-1}w_{ij}(s))| \, \|(\mathbf{w}_j(s - \xi^{-1}w_{ij}(s)) - \mathbf{w}_i(s))\|_2 ds \\ &\leq \eta + \alpha \int_0^t \sum_{j\neq i}^{N} \frac{1}{N-1} 2\theta ds \\ &\leq \eta + 2\alpha\theta \int_0^t \sum_{j\neq i}^{N} \frac{1}{N-1} ds \\ &\leq \eta + 2\alpha\theta \int_0^t ds \\ &\leq \eta + 2\alpha\theta T \leq \theta \end{aligned}$$

for T sufficiently small, more precisely for $0 < T \leq \frac{\theta-\eta}{2\alpha\theta}$. Thus, under this condition on T, if $\mathbf{w} \in V$ then $S(\mathbf{w}) \in V$ and so S maps the space V into itself.

Now, let's prove that S is a contraction. For $\mathbf{w}^{(1)}$ and $\mathbf{w}^{(2)}$ arbitrary elements of V, we have $\|S_i(\mathbf{w}^{(1)})(t) - S_i(\mathbf{w}^{(2)})(t)\|_2 = 0$, for $t \in [-J, 0]$ and $i, j \in [N]$, and

$$\begin{aligned}
\|S_i(\mathbf{w}^{(1)})(t) - S_i(\mathbf{w}^{(2)})(t)\|_2 &= \Big\|\alpha \int_0^t \sum_{j\neq i}^{N} [a_{ij}(s-\xi^{-1}w_{ij}^{(1)}(s))(\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s)) - \mathbf{w}_i^{(1)}(s))] \\
&\quad -[a_{ij}(s-\xi^{-1}w_{ij}^{(2)}(s))(\mathbf{w}_j^{(2)}(s-\xi^{-1}w_{ij}^{(2)}(s)) - \mathbf{w}_i^{(2)}(s))]ds\Big\|_2 \\
&\leq \alpha \sum_{j\neq i}^{N} \int_0^t \|[a_{ij}(s-\xi^{-1}w_{ij}^{(1)}(s))(\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s)) - \mathbf{w}_i^{(1)}(s))] \\
&\quad -[a_{ij}(s-\xi^{-1}w_{ij}^{(2)}(s))(\mathbf{w}_j^{(2)}(s-\xi^{-1}w_{ij}^{(2)}(s)) - \mathbf{w}_i^{(2)}(s))]\|_2 ds \\
&\leq \alpha \sum_{j\neq i}^{N} \int_0^t \|(a_{ij}(s-\xi^{-1}w_{ij}^{(1)}(s)) - a_{ij}(s-\xi^{-1}w_{ij}^{(2)}(s)))(\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s)) \\
&\quad +a_{ij}(s-\xi^{-1}w_{ij}^{(2)}(s))(\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s)) - \mathbf{w}_j^{(2)}(s-\xi^{-1}w_{ij}^{(2)}(s)) + \mathbf{w}_i^{(2)}(s \\
&\leq \alpha \sum_{j\neq i}^{N} \int_0^t |a_{ij}(s-\xi^{-1}w_{ij}^{(1)}(s)) - a_{ij}(s-\xi^{-1}w_{ij}^{(2)}(s))| \cdot \|(\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s) \\
&\quad +a_{ij}(s-\xi^{-1}w_{ij}^{(2)}(s))\|(\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s)) - \mathbf{w}_j^{(2)}(s-\xi^{-1}w_{ij}^{(2)}(s)) + \mathbf{w}_i^{(2)}(\\
&\leq \alpha \sum_{j\neq i}^{N} \int_0^t L\xi^{-1}|w_{ij}^{(1)}(s) - w_{ij}^{(2)}(s)|(\|\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s))\|_2 + \|\mathbf{w}_i^{(1)}(s)\|_2) \\
&\quad +\frac{1}{N-1}(\|\mathbf{w}_j^{(1)}(s-\xi^{-1}w_{ij}^{(1)}(s)) - \mathbf{w}_j^{(2)}(s-\xi^{-1}w_{ij}^{(2)}(s))\|_2 + \|\mathbf{w}_i^{(2)}(s) - \mathbf{w}_i^{(1 \\
&\leq \alpha \sum_{j\neq i}^{N} \int_0^t L\xi^{-1} 2\|\mathbf{w}^{(1)} - \mathbf{w}^{(2)}\| + \frac{1}{N-1} 2\|\mathbf{w}^{(1)} - \mathbf{w}^{(2)}\| ds \\
&\leq 2\alpha[(N-1)L\xi^{-1} + 1]T\|\mathbf{w}^{(1)} - \mathbf{w}^{(2)}\|
\end{aligned}$$

Therefore, for

$$T < \min\left\{\frac{\theta-\eta}{2\alpha\theta}, \frac{1}{2\alpha[(N-1)L\xi^{-1}+1]}\right\},$$

S maps the space V into itself and is a contraction. Hence it has a unique fixed point, which is the solution to the intial-value problem (3)-(5). This conludes the proof of the theorem. □

3. Numerical simulations

Our goal in this section is to investigate the behavior of solutions to the initial-value problem (4)-(5) numerically. We are particularly interested in illustrating and visualizing the flocking behavior, whose definition is introduced next.

Definition 1. *The solution to the delay initial-value problem (4)-(5) exhibits flocking behavior if*

$$\sup_{t\geq 0} d_{\mathbf{x}}(t) < \infty \quad \textit{and} \quad \lim_{t\to\infty} d_{\mathbf{v}}(t) = 0. \tag{10}$$

[To be completed by CP and GJ: Overview of numerical simulations, methodology, and comparative model structure.]

3.1. **Scenario 1: Baseline Consensus Under Small Delays. [To be completed by CP and GJ: Description of scenario configuration and physical setup.]**

TABLE 1. Simulation parameters for Scenario 1.

Parameter	Symbol	Value	Description
Agents	N	6	Small flock
Step size	h	0.5	Euler integration step
Interaction strength	α	0.25	Moderate coupling
Decay exponent	β	0.4	Moderate spatial decay
Fixed delay	τ	0.04	Very small (near-instantaneous)
Delay factor	τ_{factor}	0.01	Very small distance scaling
Convergence threshold	ϵ	0.01	Velocity consensus criterion

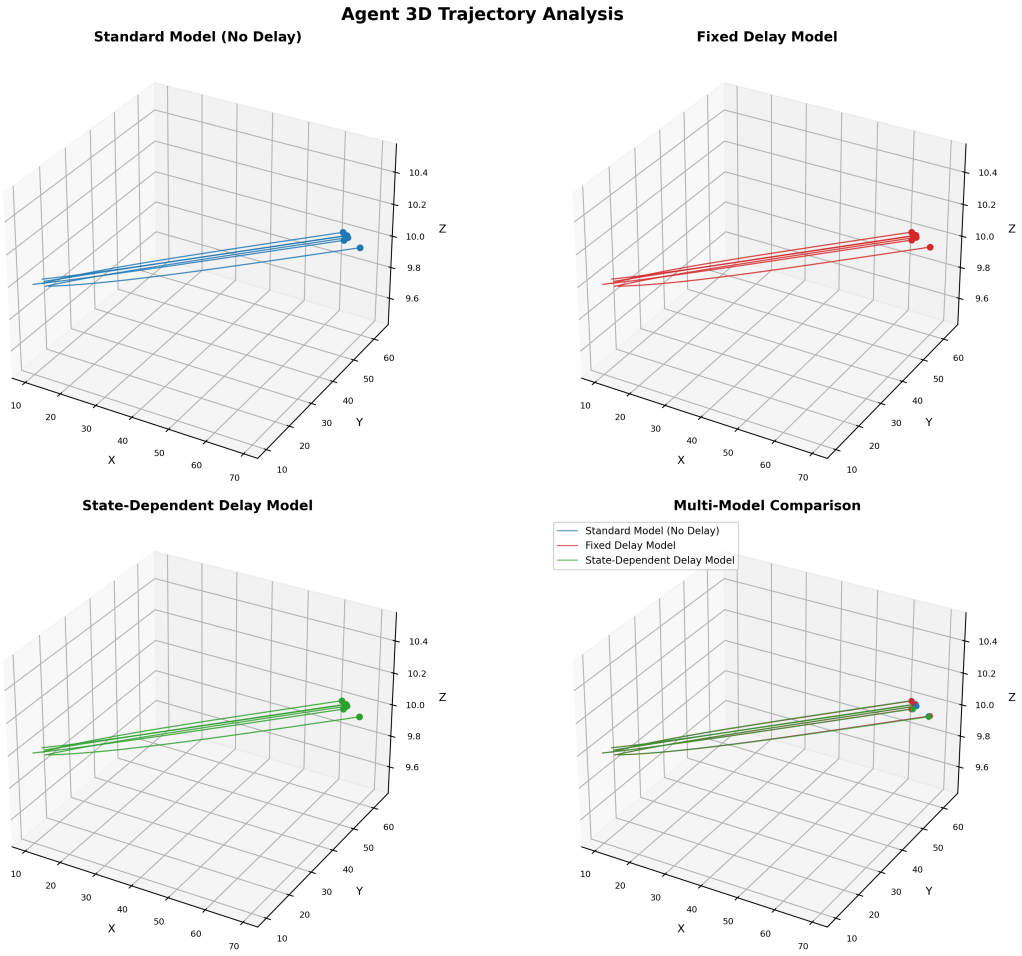


FIGURE 1. Agent 3D trajectories for Scenario 1 (top-left: standard model; top-right: fixed-delay; bottom-left: state-dependent delay; bottom-right: combined).

[To be completed by CP and GJ: Description of 3D Trajectory Plot (Figure 1).]

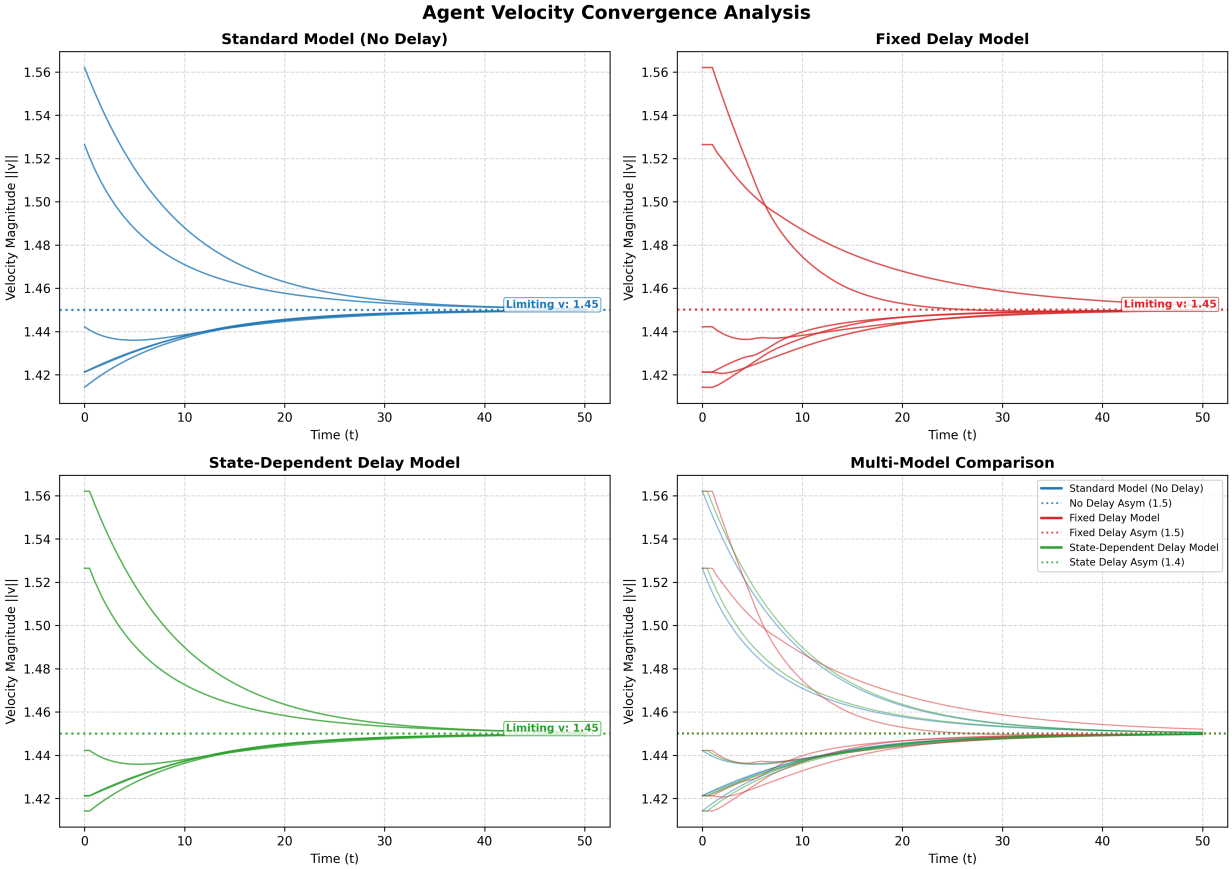


FIGURE 2. Velocity magnitude $\|\mathbf{v}_i(t)\|$ of each agent over time for Scenario 1 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined). Dotted lines mark the limiting velocity asymptotes for each flock cluster.

[To be completed by CP and GJ: Description of Velocity Convergence Plot (Figure 2).]

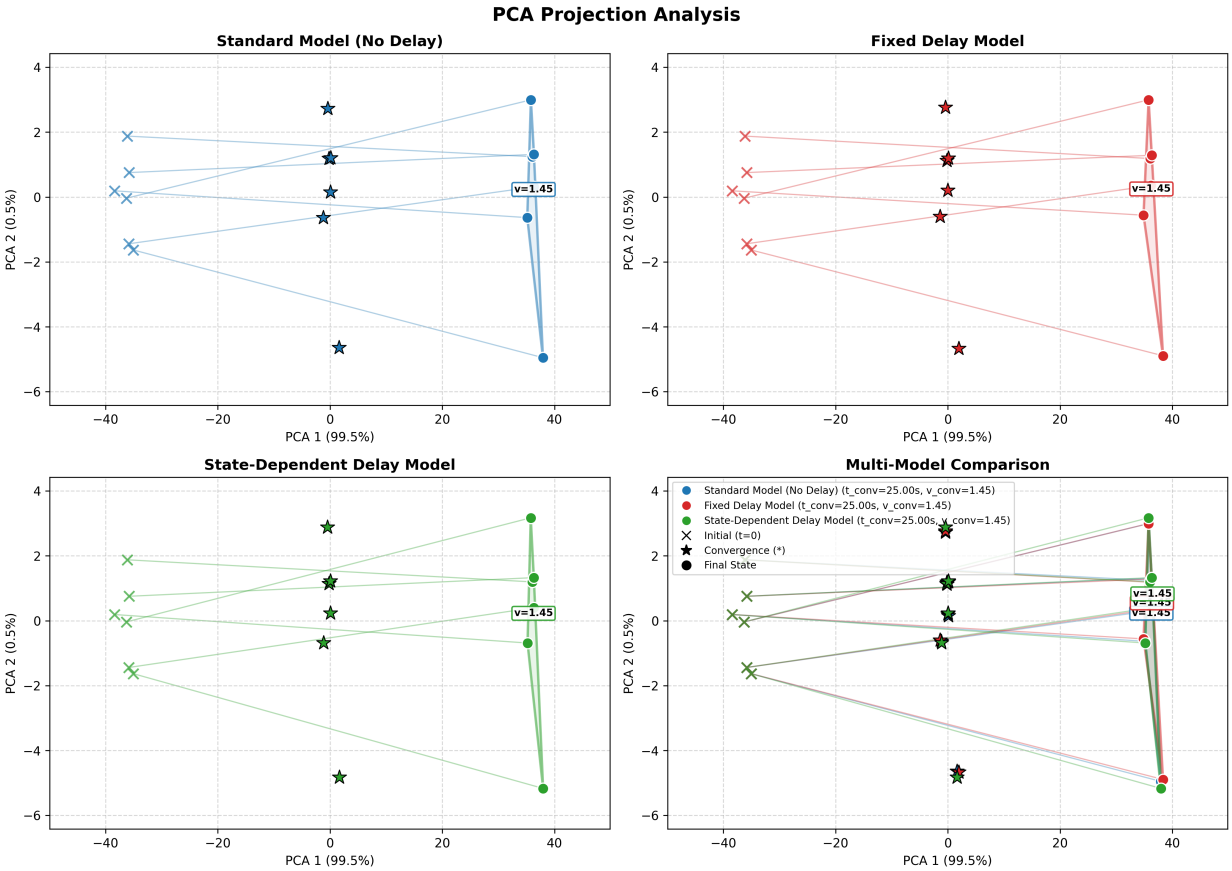


FIGURE 3. PCA projection analysis for Scenario 1 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined multi-model comparison). Crosses mark initial states ($t = 0$), stars mark convergence points, and filled circles mark terminal states with velocity cluster bounds.

[To be completed by CP and GJ: Description of PCA Plot (Figure 3), noting that groups are highlighted by same limiting velocity.]

3.2. **Scenario 2: Strong Interaction and Rapid Consensus. [To be completed by CP and GJ: Description of scenario configuration and physical setup.]**

TABLE 2. Simulation parameters for Scenario 2.

Parameter	Symbol	Value	Description
Agents	N	6	Small flock
Step size	h	0.5	Euler integration step
Interaction strength	α	0.6	High coupling
Decay exponent	β	0.2	Low decay (long-range interaction)
Fixed delay	τ	0.01	Very small
Delay factor	τ_{factor}	0.005	Very small distance scaling
Convergence threshold	ϵ	0.01	Velocity consensus criterion

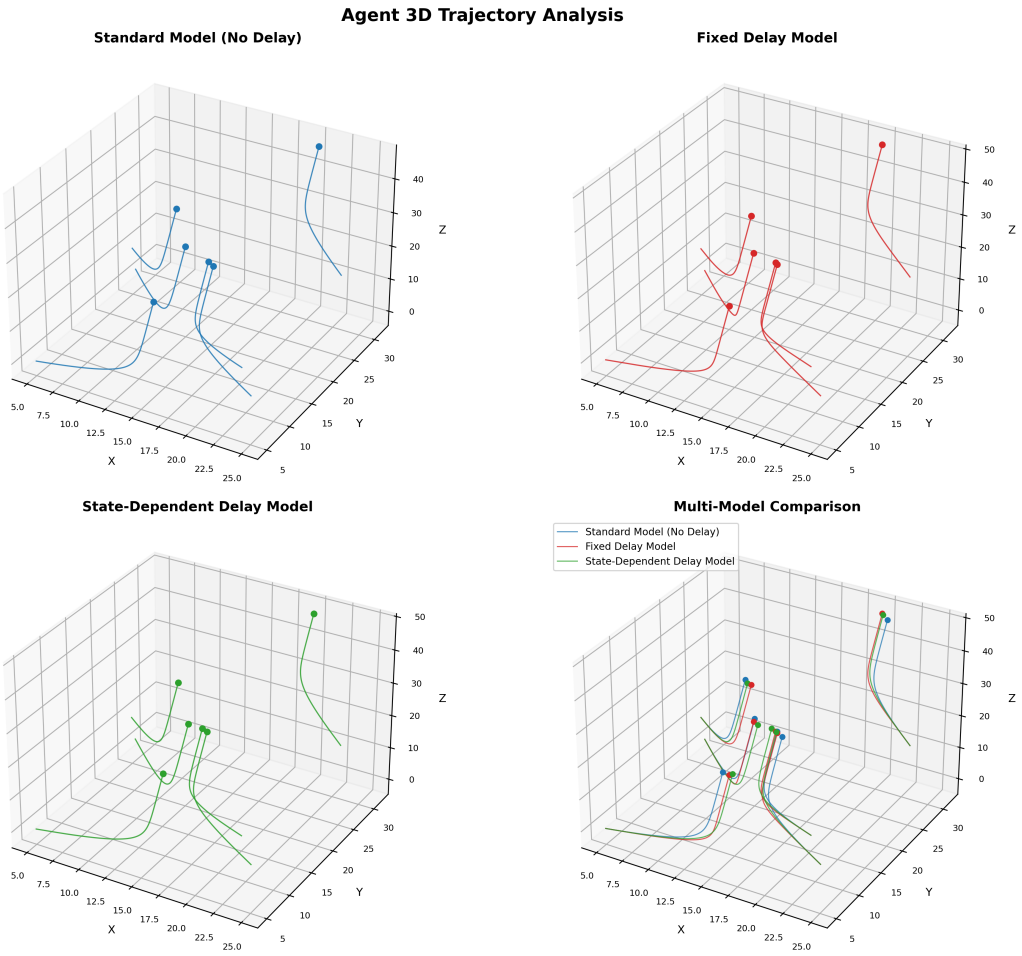


FIGURE 4. Agent 3D trajectories for Scenario 2 (top-left: standard model; top-right: fixed-delay; bottom-left: state-dependent delay; bottom-right: combined).

[To be completed by CP and GJ: Description of 3D Trajectory Plot (Figure 4).]

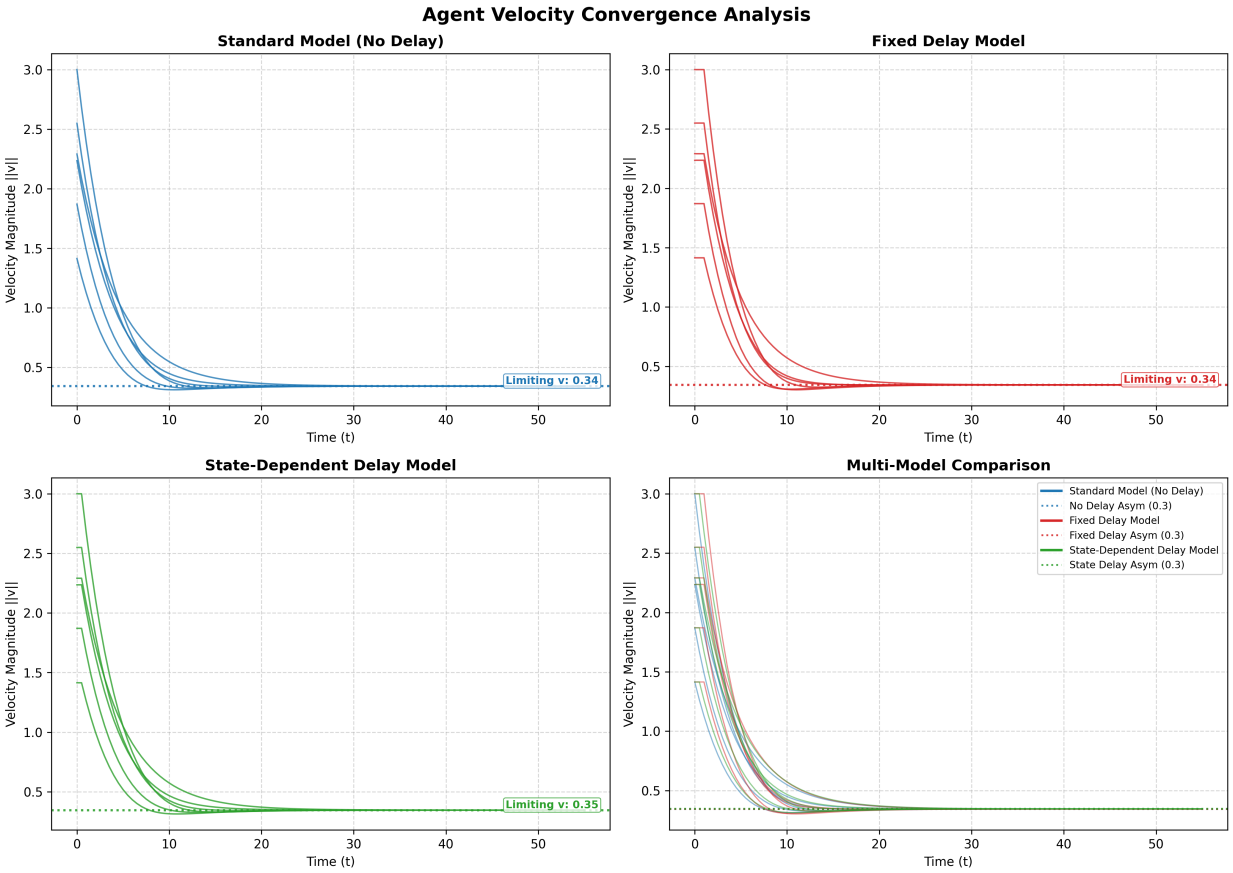


FIGURE 5. Velocity magnitude $\|\mathbf{v}_i(t)\|$ of each agent over time for Scenario 2 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined). Dotted lines mark the limiting velocity asymptotes for each flock cluster.

[To be completed by CP and GJ: Description of Velocity Convergence Plot (Figure 5).]

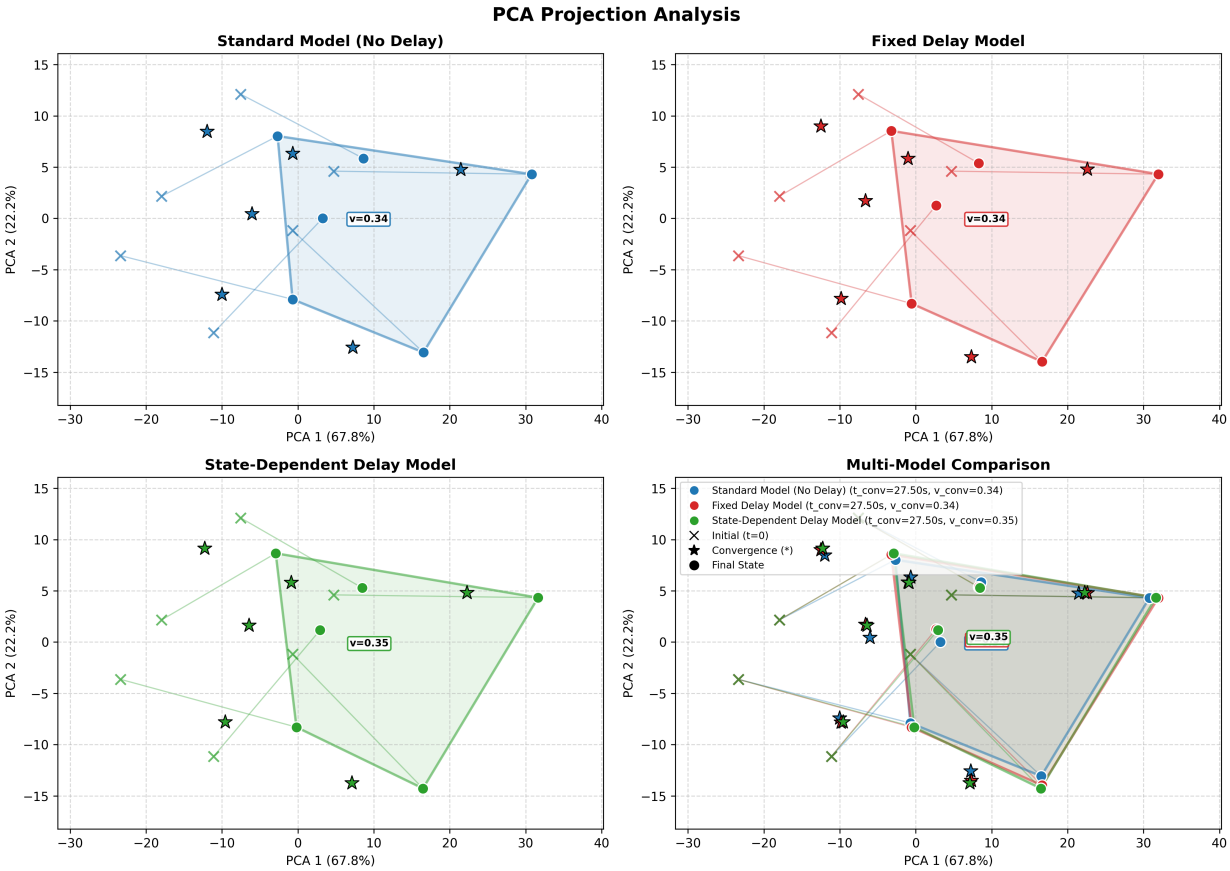


FIGURE 6. PCA projection analysis for Scenario 2 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined multi-model comparison). Crosses mark initial states ($t = 0$), stars mark convergence points, and filled circles mark terminal states with velocity cluster bounds.

[To be completed by CP and GJ: Description of PCA Plot (Figure 6), noting that groups are highlighted by same limiting velocity.]

3.3. **Scenario 3: Multi-flocking and Cluster Split. [To be completed by CP and GJ: Description of scenario configuration and physical setup.]**

TABLE 3. Simulation parameters for Scenario 3.

Parameter	Symbol	Value	Description
Agents	N	6	Two groups of 3
Step size	h	0.5	Euler integration step
Interaction strength	α	0.15	Low coupling
Decay exponent	β	0.8	High decay (short-range only)
Fixed delay	τ	0.15	Moderate
Delay factor	τ_{factor}	0.05	Moderate distance scaling
Convergence threshold	ϵ	0.01	Velocity consensus criterion

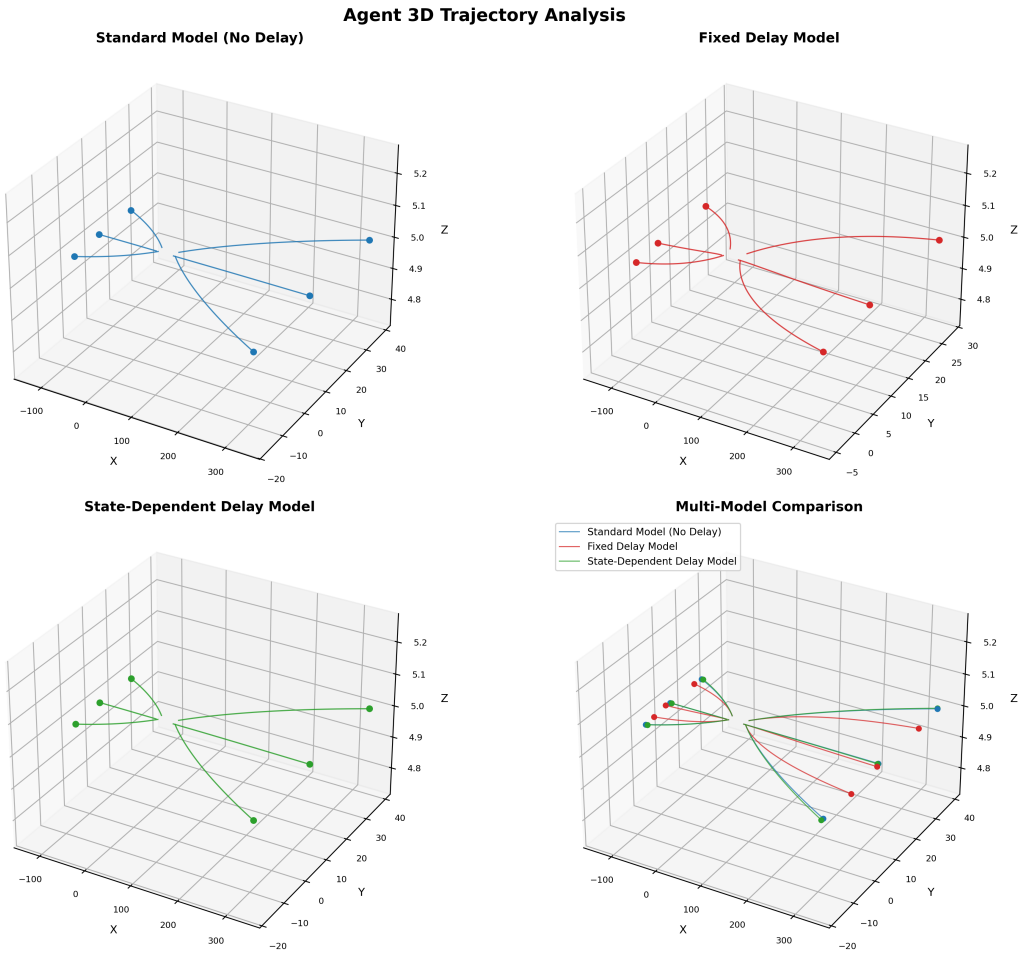


FIGURE 7. Agent 3D trajectories for Scenario 3 (top-left: standard model; top-right: fixed-delay; bottom-left: state-dependent delay; bottom-right: combined).

[To be completed by CP and GJ: Description of 3D Trajectory Plot (Figure 7).]

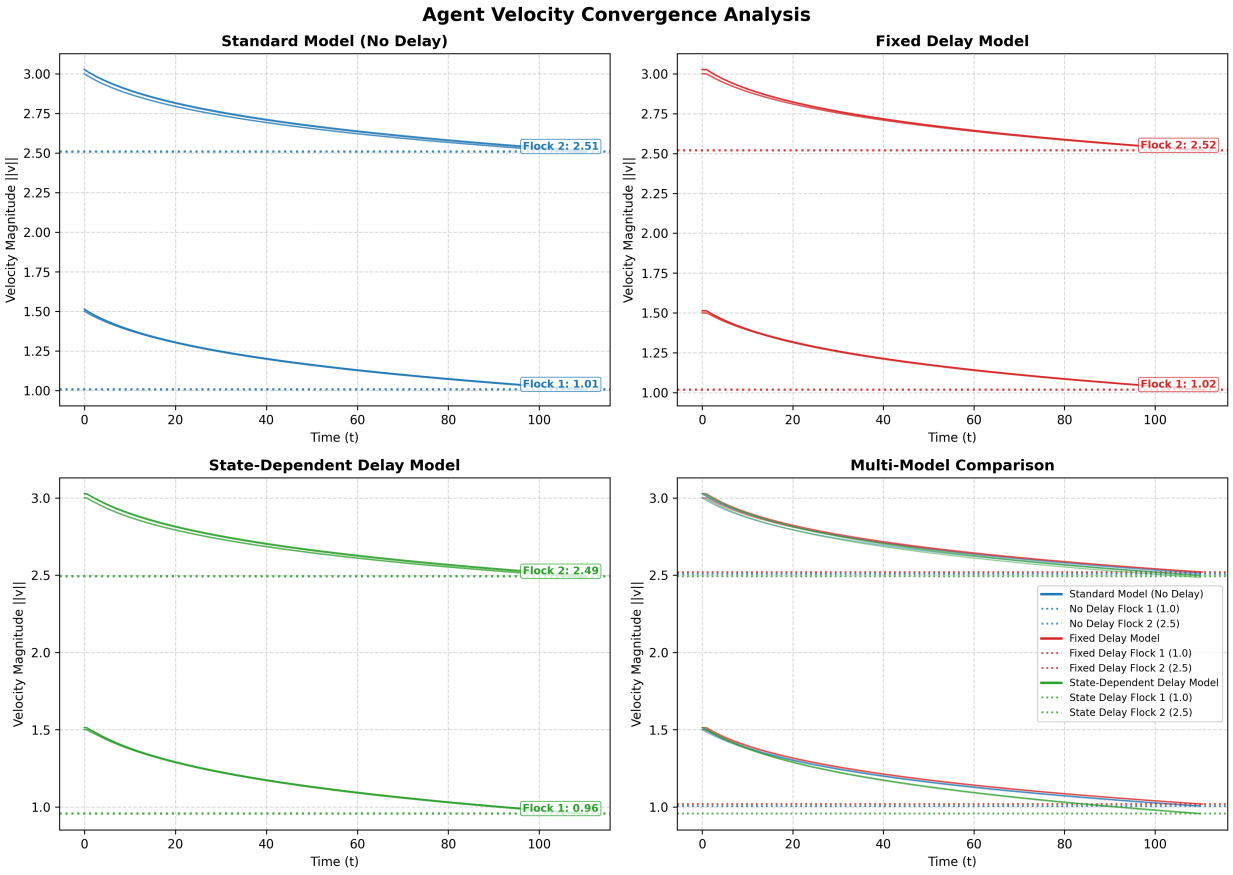


FIGURE 8. Velocity magnitude $\|\mathbf{v}_i(t)\|$ of each agent over time for Scenario 3 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined). Dotted lines mark the limiting velocity asymptotes for each flock cluster.

[To be completed by CP and GJ: Description of Velocity Convergence Plot (Figure 8).]

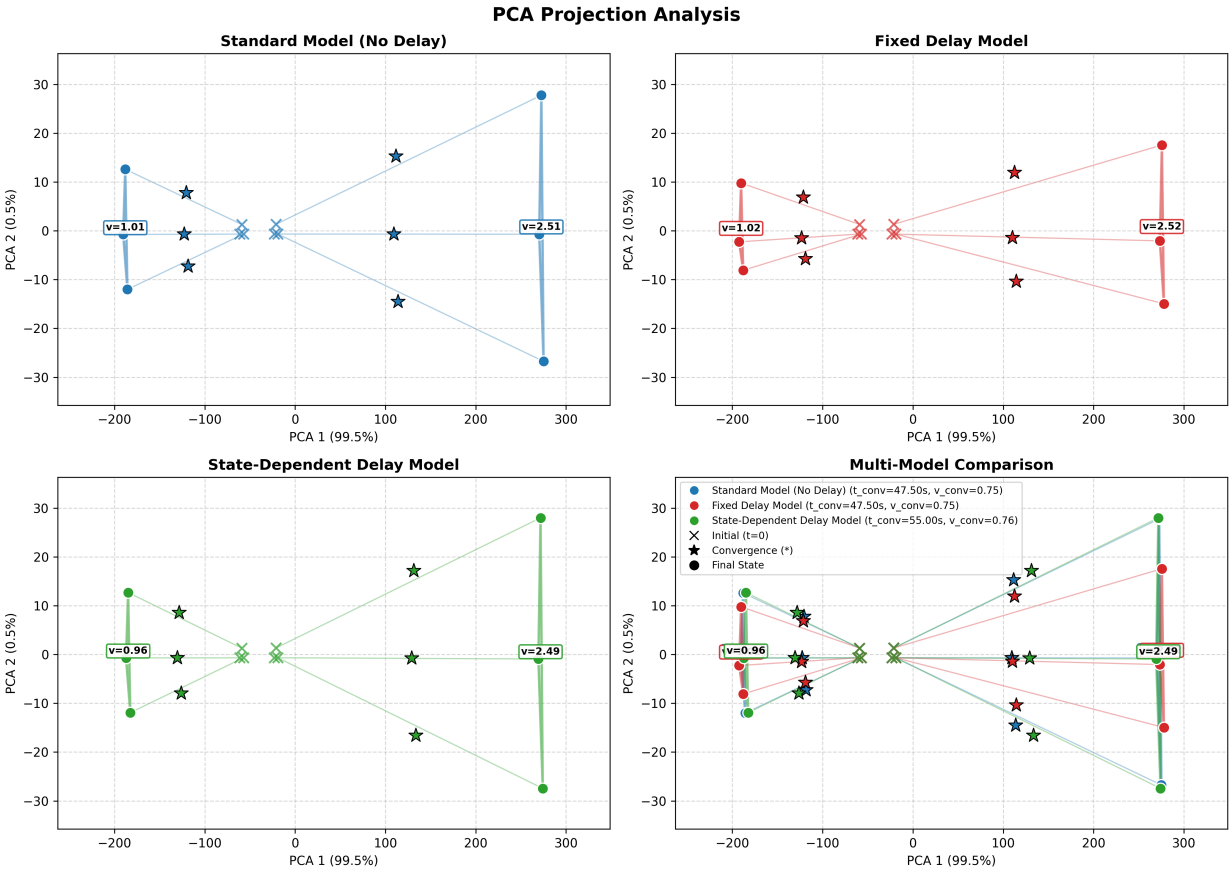


FIGURE 9. PCA projection analysis for Scenario 3 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined multi-model comparison). Crosses mark initial states ($t = 0$), stars mark convergence points, and filled circles mark terminal states with velocity cluster bounds highlighting multi-flocking behavior.

[To be completed by CP and GJ: Description of PCA Plot (Figure 9), noting that groups are highlighted by same limiting velocity to show multi-flocking causing multiple flocks with different limiting velocities.]

3.4. **Scenario 4: Large-Delay Communication and Model Divergence. [To be completed by CP and GJ: Description of scenario configuration and physical setup.]**

TABLE 4. Simulation parameters for Scenario 4.

Parameter	Symbol	Value	Description
Agents	N	10	Two groups of 5
Step size	h	0.5	Euler integration step
Interaction strength	α	0.25	Moderate coupling
Decay exponent	β	0.4	Moderate spatial decay
Fixed delay	τ	2.5	Large
Delay factor	τ_{factor}	0.15	Large distance scaling
Convergence threshold	ϵ	0.02	Velocity consensus criterion

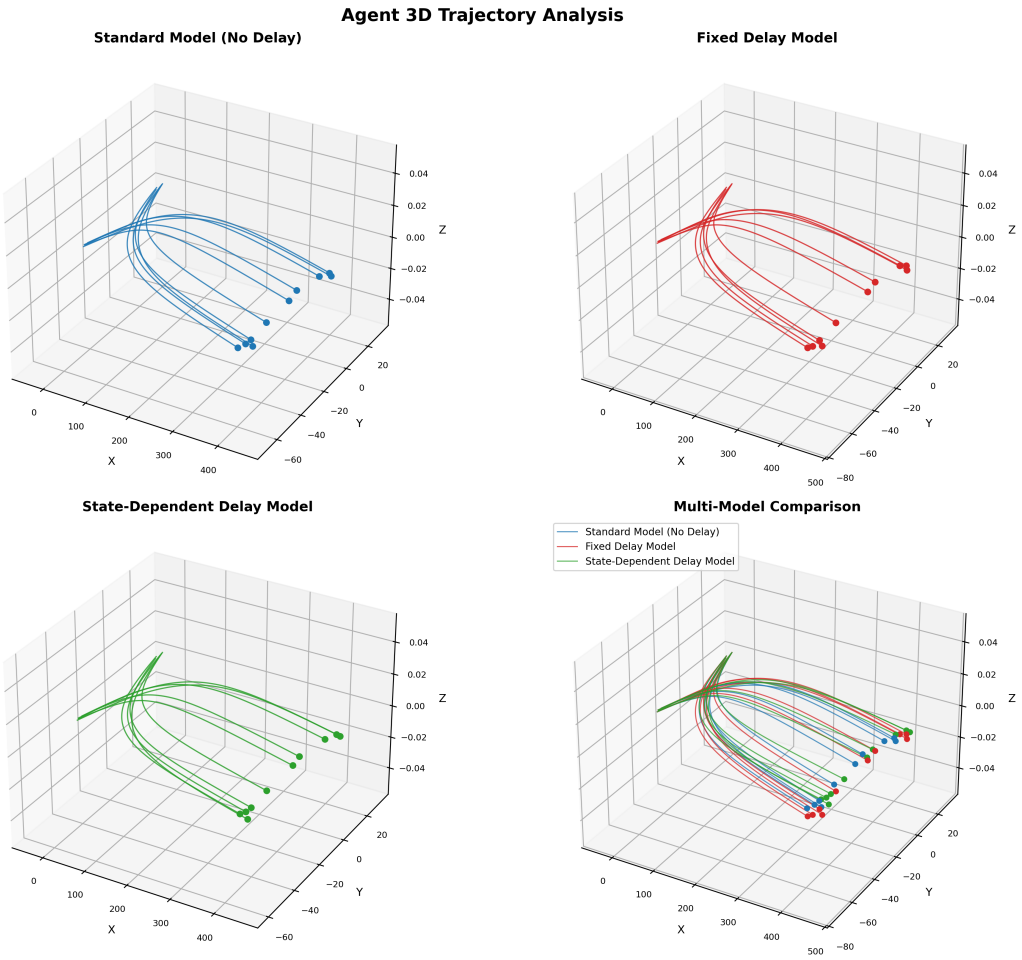


FIGURE 10. Agent 3D trajectories for Scenario 4 (top-left: standard model; top-right: fixed-delay; bottom-left: state-dependent delay; bottom-right: combined).

[To be completed by CP and GJ: Description of 3D Trajectory Plot (Figure 10).]

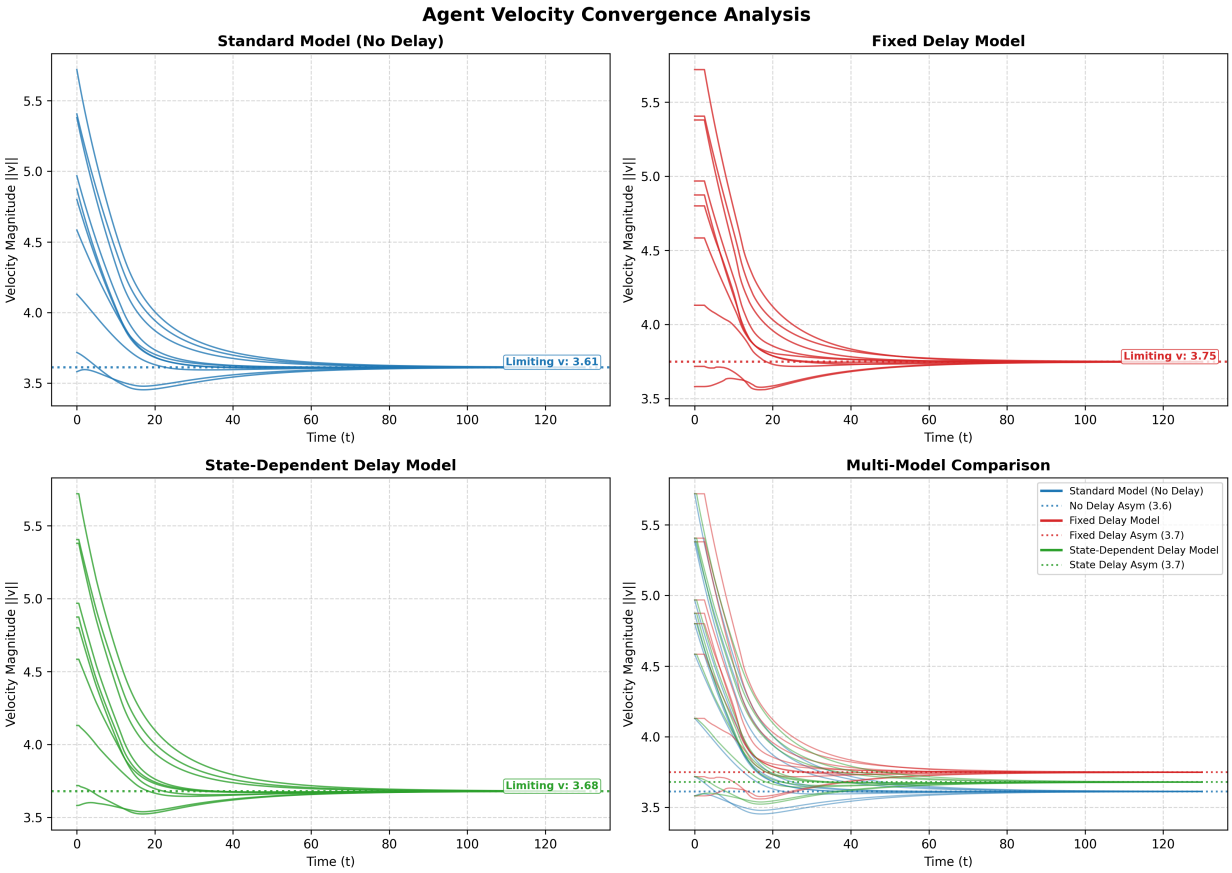


FIGURE 11. Velocity magnitude $\|\mathbf{v}_i(t)\|$ of each agent over time for Scenario 4 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined). Dotted lines mark the limiting velocity asymptotes for each flock cluster.

[To be completed by CP and GJ: Description of Velocity Convergence Plot (Figure 11).]

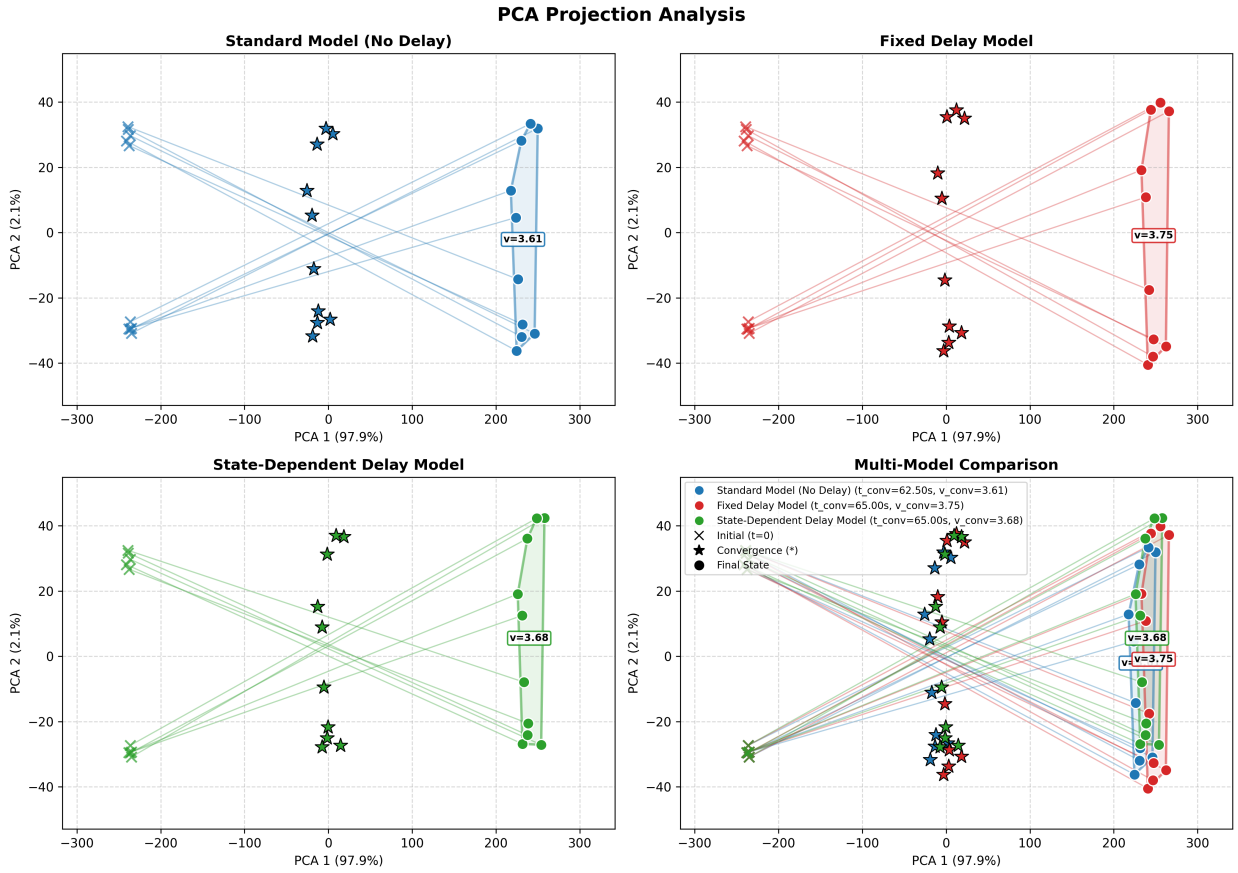


FIGURE 12. PCA projection analysis for Scenario 4 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined multi-model comparison). Crosses mark initial states ($t = 0$), stars mark convergence points, and filled circles mark terminal states with velocity cluster bounds.

[To be completed by CP and GJ: Description of PCA Plot (Figure 12), noting that groups are highlighted by same limiting velocity.]

3.5. **Scenario 5: Multi-flocking with Rapid Convergence and High Agent Count. [To be completed by CP and GJ: Description of scenario configuration and physical setup.]**

TABLE 5. Simulation parameters for Scenario 5.

Parameter	Symbol	Value	Description
Agents	N	16	Four groups of 4
Step size	h	0.5	Euler integration step
Interaction strength	α	0.08	Low coupling
Decay exponent	β	1.0	High decay (maintains cluster split)
Fixed delay	τ	0.05	Moderate delay
Delay factor	τ_{factor}	0.02	Distance-to-latency scaling
Convergence threshold	ϵ	0.005	Threshold for precise convergence

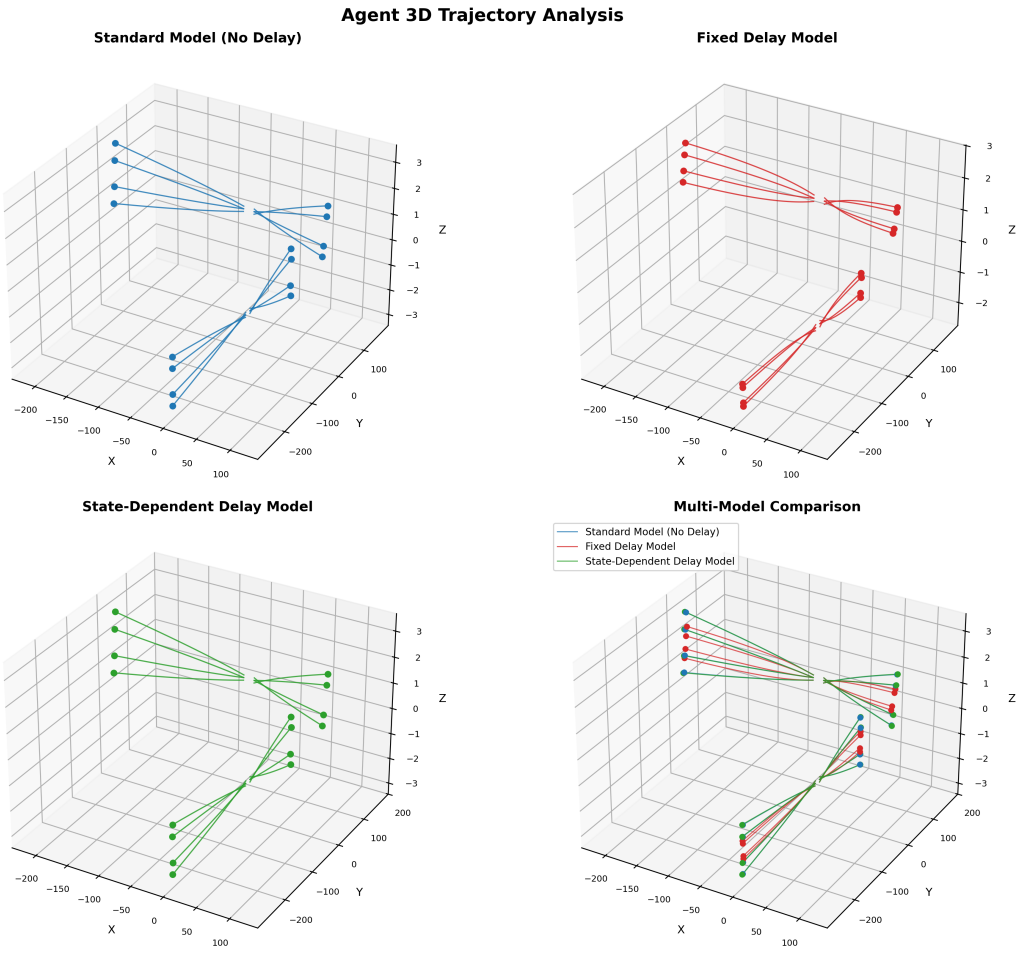


FIGURE 13. Agent 3D trajectories for Scenario 5 (top-left: standard model; top-right: fixed-delay; bottom-left: state-dependent delay; bottom-right: combined).

[To be completed by CP and GJ: Description of 3D Trajectory Plot (Figure 13).]

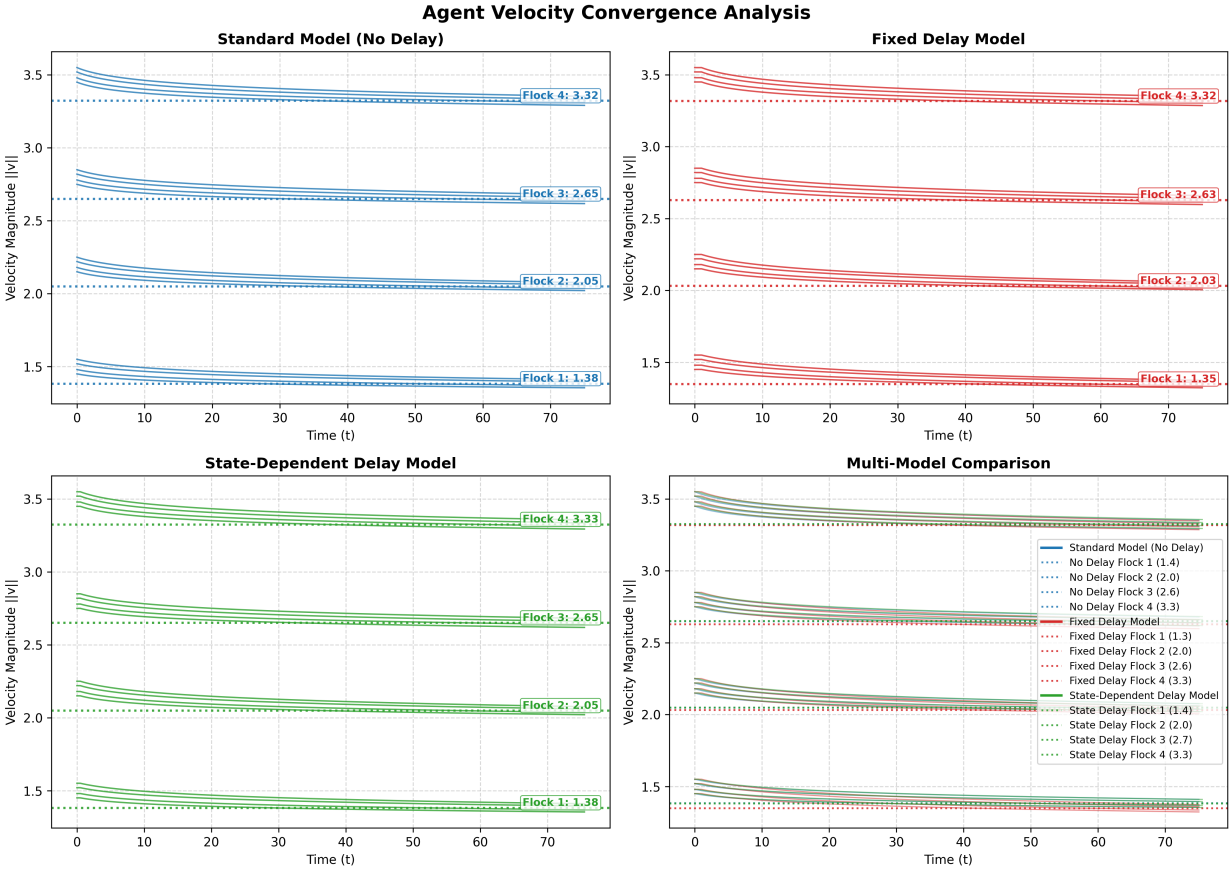


FIGURE 14. Velocity magnitude $\|\mathbf{v}_i(t)\|$ of each agent over time for Scenario 5 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined). Dotted lines mark the limiting velocity asymptotes for each flock cluster.

[To be completed by CP and GJ: Description of Velocity Convergence Plot (Figure 14).]

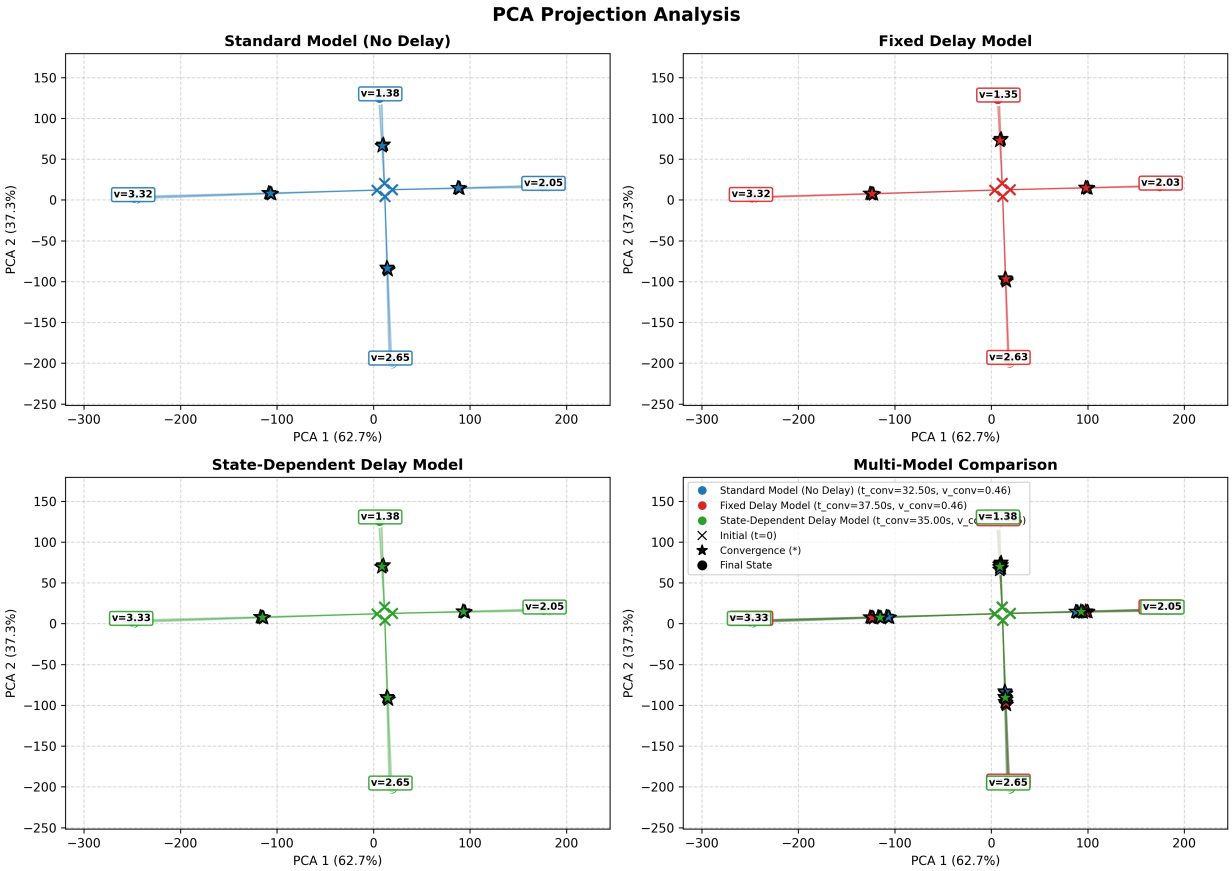


FIGURE 15. PCA projection analysis for Scenario 5 (top-left: standard model, blue; top-right: fixed-delay, red; bottom-left: state-dependent delay, green; bottom-right: combined multi-model comparison). Crosses mark initial states ($t = 0$), stars mark convergence points, and filled circles mark terminal states with velocity cluster bounds highlighting multi-flocking behavior for $N = 16$ agents.

[**To be completed by CP and GJ: Description of PCA Plot (Figure 15), noting that groups are highlighted by same limiting velocity.**]

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